MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025
SECOND SEMESTER
DSC-MG2DSCMAT100- A GATEWAY TO MATHEMATICS

SCHEME OF VALUATION
QP CODE: 25900138

Duration: 2 hours Max Marks: 70

Part A

Short Answer Type Questions

Answer any **FIVE** Questions, each question carries **2 Marks**

1. $\frac{\partial f}{\partial x} = 30x^2$ ……..…1 Mark
   $\frac{\partial f}{\partial y} = 48y^3$…… 1 Mark [U]/CO1
2. An interior point of the domain of the function $f(x, y)$ where f_x and f_y are zero or where one or both of f_x and f_y do not exist…… 2 Marks [U]/C01
3. A partition of the interval [a, b] is a collection of points $a = x_0 < x_1 < \ldots\ldots < x_n = b$ that divides [a, b] into n subintervals … 1 Mark
   A partition is regular if all the subintervals have the same length $\Delta x = \frac{b-a}{n}$ 1 Mark [A]/CO2
4. The fundamental theorem of calculus part 2: $\frac{d}{dx}\int_a^x f(t)dt = f(x)$……. 2 Marks [U]/CO2
5. m rows and n columns ……...… 2 Marks [A]/CO3
6. If the system has no solution at all ……..... 2 Marks [U]/CO3
7. $A = [a_{ij}]$, where a_{ij} is the number of edges joining ith vertex and jth vertex 2 Marks [K]/CO4
8. 14 edges (Give full marks to 16 edges also) …… 2 Marks [A]/CO4

PART B

Short Essay Type Questions

Answer any **FIVE** questions, each question carries **6 Marks**

9. $\frac{dy}{dx} = -\frac{Fx}{Fy}$ …… 1 Mark
   $Fx = e^y + y\cos xy$ …… 1 Mark
   $Fy = xe^y + x\cos xy + 1$…… 1 Mark
   $\frac{dy}{dx} = -\frac{e^y + y\cos xy}{xe^y + x\cos xy + 1}$ ……. 1 Mark
   $\frac{dy}{dx}\ at\ (0, \ln 2) = -(2 + \ln 2)$ ……..…. 2 Marks [U]/CO1

10. $\frac{\partial T}{\partial r} = \frac{\partial T}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial T}{\partial y}\frac{\partial y}{\partial r}$ …… 1 Mark

    $\frac{\partial T}{\partial \theta} = \frac{\partial T}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial T}{\partial y}\frac{\partial y}{\partial \theta}$ …… 1 Mark

$\frac{\partial T}{\partial x} = 2xy - y^3$, $\frac{\partial T}{\partial y} = x^2 - 3xy^2$ ..... 1 Mark

$\frac{\partial x}{\partial r} = \cos\theta$, $\frac{\partial y}{\partial r} = \sin\theta$, $\frac{\partial x}{\partial \theta} = -r\sin\theta$, $\frac{\partial y}{\partial \theta} = r\cos\theta$..... 1 Mark

$\frac{\partial T}{\partial r} = (2xy - y^3)\cos\theta + (x^2 - 3xy^2)\sin\theta$ ..... 1 Mark

$\frac{\partial T}{\partial \theta} = (2xy - y^3)(-r\sin\theta) + (x^2 - 3xy^2)\, r\cos\theta$ ..... 1 Mark [U]/CO1

11. $= \int_{-1}^{1}\left[4y - x^2y - \frac{y^3}{3}\right]_0^2 dx$ ..... 1 Mark

$= \int_{-1}^{1}(\frac{16}{3} - 2x^2)\, dx$ ..... 2 Marks

$= \left[\frac{16x}{3} - \frac{2x^3}{3}\right]_{-1}^{1}$ ..... 1 Mark

$= \frac{28}{3}$ ..... 2 Marks [A]/CO2

12. $= \int_0^{\pi/3}[\sin 2x \frac{\sin 3y}{3}]_0^{\pi/6}\, dx$ ..... 1 Mark

$= \int_0^{\pi/3} \frac{\sin 2x}{3}\, dx$ ..... 2 Marks

$= [\frac{-\cos 2}{6}]_0^{\pi/3}$ ..... 1 Mark

$= \frac{1}{4}$ ..... 2 Marks [A]/CO2

13. Writing augmented matrix ..... 1 Mark

Obtaining $\left[\begin{array}{ccc|c} 1 & -1 & 1 & 4 \\ 0 & 1 & 3 & -5 \\ 0 & 0 & 1 & -1 \end{array}\right]$ or any other equivalent ..... 2 Marks

Observing $r = 3 = n$ ..... 1 Mark

Concluding the system has exactly one solution..... 2 Marks [A]/CO3

14. Writing augmented matrix ..... 1 Mark

Obtaining $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 2 & 0 \end{array}\right]$ or equivalent ..... 2 Marks

Writing equations $x + y + z = 3, y + 2z = 1, 2z = 0$ ..... 1 Mark

Getting $x = 2, y = 1, z = 0$..... 2 Marks [A]/CO3

15. Number of vertices $= r + s$ ..... 3 Marks

Number of edges $= r * s$ ..... 3 Marks [U]/CO4

16. Correct proof ..... 6 Marks [An]/CO4

Part C

Essay Type Questions

Answer any **THREE** Questions, Each Question carries **10 Marks**

17. $f_x = 2x$, $f_y = 2y$ ..... 2 Marks

Finding points $(0,0), (0,2), (1,0)$..... 2 Marks

Evaluating $f(x,y)$ and its derivative on each side and hence getting $\left(\frac{4}{5},\frac{2}{5}\right)$… 2 Marks

Absolute Maximum 4 at (0,2) …… 2 Marks

Absolute Minimum 0 at (0,0) …… … 2 Marks [U]/CO1

18. (a) Statement $\int_a^b f(x)\,\mathrm{dx} = \mathrm{F(b)} - \mathrm{F(a)}$ …… 3 Marks

(b) $\int_0^4 f(x)\ \mathrm{dx} = \int_0^1 \sqrt{x}\ \mathrm{dx} + \int_1^4 \frac{1}{x^2}\,\mathrm{dx}$ …….. 2 Marks

$= \left[\frac{x^{3/2}}{3/2}\right]_0^1 + \left[\frac{-1}{x}\right]_1^4$ …… 2 Marks

Substituting and getting value $= \frac{17}{12}$ ……..…….. 3 Marks [A]/CO2

19. $\int_0^1 5\,dx - \int_0^1 3\sqrt{1-x^2\,dx}$ ……. 2 Marks

Evaluating first integral as 5 …… 2 Marks

Evaluating second integral as $\frac{3\pi}{4}$ ……. 5 Marks

Final answer $5 - \frac{3\pi}{4}$ ……..…. 1 Mark [A]/CO2

20. (a) Writing augmented matrix and reducing …… 2 Marks

Writing equations $2x + 5y = 2,\ -13\,y = 26$ ……. 1 Mark

$x = 6, y = -2$ ……. 1 Mark

(b) Writing augmented matrix …… 1 Mark

Obtaining $\left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 10 & 25 & 90 \\ 0 & 0 & -95 & -190 \\ 0 & 0 & 0 & 0 \end{array}\right]$ or equivalent …… 2 Marks

Writing equations $x - y + z = 0, 10y + 25z = 90, -95z = -190$ …. 1 Mark

Getting $x = 2, y = 4, z = 2$……..……..……..…. 2 Marks [A]/CO3

21. (a) Definition of Edge deleted with example …… 2 Marks

Definition of Vertex deleted with example ……...……. 2 Marks

(b) Drawing C_6 ……. 1 Mark

Graph of C_6 – v..…….…… 2 Marks

Graph of C_6 –e ……..……...…….... 2 Marks

(c) Yes ……...……. 1 Mark [U]/CO4

22. (a) C_8 with loop edges at vertices A, C and F ……...…. 7 Marks

(b) AA, CC and FF ……..…… 3 Marks [U]/CO4

NB: Stage wise marks for any correct alternate method can be fixed by the examiner